

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Senior AI and software engineering professional with **5+ years of experience** architecting and shipping production-grade software systems using Python, TypeScript, REST APIs, microservices, distributed systems, cloud infrastructure, and modern AI/LLM technologies.
- Hands-on experience building **production AI agent systems, LLM-powered products, multi-component agent pipelines, multi-step reasoning workflows, tool integrations, and orchestration/runtime infrastructure** using modern LLM APIs and agent frameworks.
- Strong backend engineering foundation in **Python, FastAPI, Flask, service architecture, distributed systems, API design, asynchronous processing, event-driven patterns, data pipelines, and cloud-native application development**.
- Experienced implementing **agent observability and evaluation** through tracing, automated evals, quality regression detection, drift detection, latency monitoring, cost analysis, model-output analysis, incident response, and production reliability practices.
- Owned complex technical initiatives **end-to-end from requirements and architecture through implementation, production deployment, monitoring, progressive rollout, and operational support**, collaborating closely with product and engineering stakeholders.
- Applied **AI-native and agentic development practices**, including AI-assisted coding, reusable agent components, plugin and extension architectures, tool protocols, managed agent runtimes, and rapid development workflows that increase engineering velocity.
- Experienced contributing to **RFCs, ADRs, cross-cutting design reviews, technical mentoring, feature-flagged releases, phased rollouts, blue/green deployments, security, compliance, and regulated-domain applications**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Backend Engineering:** Python, TypeScript, JavaScript, Java, C#, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Service Architecture, Async Processing, Event-Driven Architecture.
- **AI Agents & LLM Engineering:** LLM APIs, AI Agents, Agentic Systems, Agent SDKs, Multi-Step Reasoning, Tool Integration, Function Calling, LangGraph, CrewAI, LlamaIndex, MCP, RAG, Prompt Engineering, Orchestration Patterns.
- **Agent Runtime & Platforms:** AWS Bedrock, Bedrock AgentCore, Palantir Foundry/AIP, OpenAI, Anthropic Claude, Google Gemini, Managed Agent Runtime Infrastructure, Plugin Systems, Extension Systems, Integration Ecosystems.
- **Evaluation & Observability:** Distributed Tracing, Automated Evals, Agent Evaluation, Quality Regression Detection, Drift Detection, Model Monitoring, Latency Monitoring, Cost Monitoring, Structured Logging, Incident Response, Production Diagnostics.
- **Cloud & Data:** AWS, Azure, AWS Lambda, S3, IAM, CloudWatch, Docker, Terraform, SQL, PostgreSQL, MySQL, PySpark, ETL/ELT, Vector Databases, Data Pipelines, Cloud Infrastructure.
- **Production Delivery:** CI/CD, Git, Azure DevOps, Jenkins, GitLab CI/CD, Feature Flags, Phased Rollouts, Blue/Green Deployment, Progressive Delivery, Release Automation, Production Reliability, Rollback Strategies.
- **Architecture & Leadership:** RFCs, ADRs, System Design, Architecture Reviews, Technical Decision-Making, Reusable Platforms, API Design, Security, Compliance, Mentoring, Technical Leadership, AI-Native Development Practices.

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Architected and delivered **production AI agent systems** using Python, FastAPI, Palantir AIP, modern LLM APIs, agent SDKs, RAG, tool integrations, and multi-step reasoning workflows for complex insurance and regulated enterprise operations.
- Designed multi-component agent pipelines that integrated **LLMs, enterprise APIs, retrieval systems, structured data, workflow tools, and existing platform services**, applying distributed-system and service-architecture patterns for scalable production deployments.
- Built reusable agent orchestration and runtime components using **LangGraph, MCP-style tool integration, asynchronous Python services, plugin architectures, event-driven patterns, and managed cloud infrastructure** to support reliable agent execution.
- Implemented comprehensive **agent observability and evaluation pipelines** using distributed tracing, automated evals, quality regression detection, drift monitoring, latency metrics, cost analysis, structured logging, and production health dashboards.
- Owned technical initiatives end-to-end from **requirements analysis, architecture design, RFC/ADR decisions, implementation, production deployment, incident response, and ongoing optimization**, coordinating closely with product and engineering peers.
- Applied progressive delivery strategies including **feature-flagged releases, phased rollouts, blue/green deployments, early-stage quality signals, CI/CD, Docker, Terraform, and rollback mechanisms** to improve release velocity and platform stability.
- Mentored engineers on **AI-native development, agentic coding workflows, prompt and tool design, debugging distributed agent behavior, production reliability, and reusable engineering patterns**, while contributing shared components and architectural guidance across teams.

Environments: Python, FastAPI, Palantir Foundry/AIP, OpenAI, Anthropic Claude, LangGraph, MCP, RAG, Agent SDKs, Tool Calling, AWS, Bedrock AgentCore, REST APIs, Distributed Systems, Docker, Terraform, CI/CD, Feature Flags, Blue/Green Deployment, Tracing, Automated Evals, Drift Detection, Observability.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed production healthcare platforms using **Python, FastAPI, REST APIs, AWS, SQL, microservices, and distributed-system architecture**, supporting secure and highly available workflows within a regulated enterprise environment.
- Built **LLM-powered applications and agentic workflows** using modern LLM APIs, LangGraph patterns, RAG, embeddings, vector databases, tool calling, multi-step reasoning, and integration with enterprise healthcare services.
- Developed service and orchestration layers integrating **AI agents, APIs, backend platforms, data sources, and cloud services**, applying asynchronous processing, event-driven patterns, resilient API design, and reusable integration components.
- Implemented production observability through **distributed tracing, structured logging, automated evaluation pipelines, latency monitoring, output-quality checks, regression detection, CloudWatch, and Splunk**, improving reliability and incident diagnostics.
- Owned application components from **technical design and architecture decisions through development, deployment, monitoring, incident response, and operational optimization**, collaborating with product managers, architects, and engineering teams.
- Applied progressive release practices including **feature flags, phased deployments, CI/CD, Docker, deployment validation, rollback strategies, and environment-specific configuration** to increase release stability and delivery velocity.
- Created reusable **Python libraries, API abstractions, agent components, architectural documentation, runbooks, and integration patterns** while sharing engineering practices with distributed development teams.

Environments: Python, FastAPI, OpenAI/LLM APIs, LangGraph Concepts, RAG, Vector Databases, REST APIs, Microservices, AWS, AWS Lambda, CloudWatch, SQL, Docker, CI/CD, Feature Flags, Distributed Tracing, Automated Evals, Splunk, Incident Response.

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Architected and developed enterprise backend platforms using **Python, FastAPI, Flask, REST APIs, SQL, microservices, and distributed-system design**, delivering reliable services across complex business applications.
- Built reusable service components for **workflow orchestration, asynchronous processing, API integrations, plugin-style extensions, business-rule execution, and event-driven communication** across interconnected enterprise systems.
- Designed scalable data-processing pipelines using **Python, PySpark, SQL, ETL/ELT, PostgreSQL, MySQL, and Oracle**, supporting high-volume application workloads and downstream operational services.
- Implemented production reliability and observability capabilities using **structured logging, tracing, latency metrics, exception monitoring, health checks, incident diagnostics, and operational dashboards** across distributed services.
- Owned backend components through **requirements analysis, system design, API contracts, implementation, deployment, production monitoring, troubleshooting, and optimization**, applying strong service-architecture fundamentals.
- Contributed to **architecture reviews, RFC-style technical discussions, design trade-offs, CI/CD, Docker, Git, Jenkins, phased deployments, and release-management practices** to improve maintainability and delivery consistency.
- Shared technical expertise through **code reviews, reusable libraries, architectural guidance, technical documentation, operational runbooks, and mentoring**, strengthening engineering standards across distributed development teams.

Environments: Python, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, PySpark, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Docker, Jenkins, Git, CI/CD, Structured Logging, Tracing, Incident Diagnostics, Cloud-Native Architecture.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed production backend applications using **Python, Flask, REST APIs, SQL, and microservices architecture**, supporting distributed enterprise platforms and business-critical operational workflows.
- Built reusable backend components for **service orchestration, data transformation, workflow execution, API communication, asynchronous processing, configuration management, and extension-oriented application patterns**.
- Engineered integrations between **backend services, relational databases, third-party platforms, enterprise APIs, and event-driven workflows**, applying clear API contracts and resilient service boundaries.
- Implemented production engineering capabilities including **structured logging, health monitoring, exception tracking, deployment diagnostics, service metrics, and incident troubleshooting** across distributed application components.
- Owned assigned services from **technical requirements and design through implementation, deployment, production support, issue resolution, and iterative enhancement**, collaborating closely with architects and senior engineers.
- Supported reliable delivery using **Git, Docker, CI/CD, Linux, environment configuration, release automation, rollback procedures, and secure API integration patterns** across enterprise deployment environments.
- Produced reusable **service modules, API patterns, technical documentation, deployment guides, troubleshooting procedures, and knowledge-transfer materials**, enabling distributed teams to maintain and extend delivered systems.

Environments: Python, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Distributed Systems, Event-Driven Architecture, Docker, Git, CI/CD, Linux, API Security, Structured Logging, Monitoring, Incident Response, Cloud Concepts.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.